PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Probiotics for the prevention of antibiotic-associated diarrhoea: A systematic review and meta-analysis
AUTHORS	Goodman, Clare; Keating, Georgia; Georgousopoulou, Ekavi; Hespe, Charlotte; Levett, Kate

VERSION 1 – REVIEW

REVIEWER	SZAJEWSKA, Hania The Medical University of Warsaw, Dept of Paediatrics
REVIEW RETURNED	29-Oct-2020

GENERAL COMMENTS	The manuscript 'Probiotics for the prevention of antibiotic-associated diarrhoea: A systematic review and meta-analysis' aims to update current evidence on this topic. The research question is clear and clinically important. The authors analysed a substantial amount of evidence and provided clear conclusions. There are, however, numerous issues which negatively impact the manuscript's credibility and consistency. General comments 1. The authors decided to choose statistical model (i.e. fixed vs. random effects) depending on heterogeneity of the results. However, Cochrane Handbook clearly states that 'The choice between a fixed-effect and a random-effects meta-analysis should never be made on the basis of a statistical test for heterogeneity' (chapter 10.10.4). 2. The authors state, that their main outcome was "incidence of AAD, classified as three or more episodes of loose stools (5-7 Bristol Stool Scale), in any 24-hour period during the antibiotic treatment or follow-up phases." Please clarify whether all included trials reported the outcome as stated above, including the use of Bristol Stool Scale. At least one study documented that there is a proven, large outcome heterogeneity in studies on AAD [doi: 10.1371/journal.pone.0228824]. 3. The authors have chosen odds ratios as a method of presenting intervention effects. The Cochrane Handbook states that odds and odds ratios are more difficult to interpret for a reader. Please consider presenting the results as risk ratios. 4. Throughout the manuscript, the terms 'diarrhea' and 'AAD' are used interchangeably. Please standardize the terminology in this respect. 5. The term 'flora' used throughout the manuscript is historical, and should be replaced by 'microbiota'. Also, C. difficile no longer belongs to Clostridium genus, but to Clostridioides, and should be renamed in the manuscript accordingly. 6. Throughout the manuscript, the authors refer to Joshua Goldenberg's 2013 systematic review as the 'most recent'

	Cochrane review. However, there was an update of this work by the same author in 2017. Please revise. 7. The database search was performed over a year ago. A search update is highly recommended. Section-specific comments Abstract: 8. "…co-administration of probiotics reduces the risk of antibiotic-associated diarrhoea in adults by 46% (Odds ratio (OR)=0.54 [95% CI: 0.44, 0.67], p<0.00001)" – risk and odds are two entirely different things. This sentence needs rephrasing. 9. "Probiotics are non-pathogenic live micro-organisms, thought to improve the microbial balance of the host and reduce colonisation by pathogenic bacteria." – the definition presented by the authors is different from the widely adopted ISAPP definition (https://isappscience.org/for-scientists/resources/probiotics/). Please revise in the Abstract and Introduction. 10. 'Design: Systematic review and meta-analysis.' Please revise to precisely state 'systematic review and meta-analysis of randomized-controlled trials'. Introduction: 11. "It is estimated that the increase in hospital admission due to AAD is 3.6 to 21.3 days" – hospital admissions cannot be measured in days, in contrast to hospital stay. Please revise. 12. "They can be delivered orally, adhere to target gastrointestinal epithelium and are stable in acid and bile (Dunne et al. 2001)." – this statement seems to be a slight oversimplification of our current understanding of probiotics' mechanism of action. In fact, in many cases this mechanism is unknown. According to ISAPP, 'In some cases, probiotic mechanisms of action are known; in other cases, they are not even though a health benefit has been demonstrated. A common misconception is that probiotics need to alter the gut microbiota in order to be effective. In fact, probiotics are not known to take up permanent residence in the gut despite documented health benefits.' 13. 'One of the most expensive probiotics in Australia costs less than four dollars daily (VSL#3 AU$3.96 per day, available from NourishMeOrganics.com.au, April 2020).' – usually, use of commercial product names is discouraged in articles. Please refer to BMJ Open author guidelines on this subject. Methods: 14. The authors state, that "a sensitivity analysis was performed". However, it is not described what kind of a sensitivity analysis it precisely was. Please clarify. 15. "Randomised controlled trials (RCTs) comparing probiotic use (any strain, dose or formulation) to placebo, alternative dose, alternative probiotic strain, or no treatment, for the prevention of diarrhoea in adults receiving antibiotic therapy were eligible for inclusion in this review." There is a discrepancy between this part and the abstract, which states that only 'placebo or no treatment control' will be taken into an account. Please clarify. Results: 16. "A total of 346 articles were identified in the initial search, of which 53 were eligible for fulltext review, and 37 of those articles were eligible for inclusion in the study." – if so, this search strategy has impressive specificity, considering that in recent Cochrane review mentioned previously, as much as 1351 studies were identified from the primary electronic databases. This surprisingly low number should be at least commented on in the discussion

	17. There are some missing data in Table 1, e.g. in column 2 of references 31-33, in column 5 of reference 32. 18. “All studies reported the outcome of interest; incidence of AAD as measured by three or more occurrences of diarrhoea in a 24-hour period”. Firstly, I think that this sentence should read “occurences of loose stool” rather than “occurences of diarrhoea”. Secondly, in reference to my previous comment – did really all those authors use Bristol Stool Scale? Haven’t you identified at least one trial which used different definition of diarrhoea, or didn’t use any definition at all? Or perhaps, did you exclude trials which didn’t provide a precise definition of diarrhoea? 19. “Overall 51% of studies were rated as having a low risk of bias, while 49% were at unclear risk of bias.” What were the criteria for a study to be ranked as “low” or “high” risk of bias? 20. “…demonstrated a statistically significant reduction in the incidence of AAD by 46% (Odds ratio (OR)=0.54 [95% CI: 0.44, 0.67], p<0.00001).” – similar comment as before – incidence and odds are two different things. Please revise throughout the manuscript for consistency. 21. “This suggests that the number needed to treat (NNT), to prevent one case of AAD would be 19 patients.” – how was NNT calculated? On the basis of OR? If so, then according to Cochrane Handbook, the authors should provide an information and rationale on the ‘assumed comparator risk’ (chapter 15.4.4.). 22. “On inspection, the funnel plot of the complete case analysis (Figure 3) shows little evidence of publication bias. This suggests that our results are robust to theoretical large future trials with results in the opposite direction (i.e. that probiotics increase the risk of AAD rather than decreasing it).” – I don’t understand, how low risk of publication bias translates to robustness of results to future studies with different results. Also, visually, the funnel plot is not necessarily symmetric – there are six small studies (SE>1) with results more favourable towards probiotics compared to only 2 small studies on the right side of the plot. Did you consider performing tests for asymmetry? 23. “No serious adverse events were reported in any of the included studies” – it would be also useful to include information, how many studies reported adverse events at all. Discussion 24. The results should be briefly recapped rather than rewritten completely with p-values, confidence intervals etc. Please revise. 25. Some information regarding the results is presented for the first time, eg. “we found significant heterogeneity between study designs and population, severity of illness of the study group, as well as probiotic species used, dose, timing and duration of probiotic administration.”, or “In the five RCTs investigating adjunct probiotic use with antibiotics for H.pylori treatment we found a reduction in diarrhoeal side-effects (OR=0.43, [95% CI: 0.27, 0.69], p<0.001).”. Data like this should be introduced in the “results” section, and only recapped or commented on in the discussion. 26. “One study [62] compared two strains of L. casei, Actimel® (L. casei DN-114001) and Yakult® (L. casei Shirota), for the prevention of AAD. In this study L. casei DN-114001 was significantly more effective than L. casei Shirota , however, we were unable to include this data in our review as it was not randomised.” – I don’t understand, why this specific study is mentioned? I don’t think that it contributes much to the discussion. I guess there were more studies excluded for similar reasons, so why mention this one?

	27. “A limitation of this study is that the only outcome measured was incidence of diarrhoea. This review’s utility in clinical practice may have been improved by also analysing duration of AAD.” – this is a very important point, however I would also add for consideration: diarrhea severity, need for hospitalization due to diarrhoea, quality of life and other outcomes of real-life importance. Considering the fact, that the definition of diarrhoea in the included trials was ‘3 or more loose stools in 24 h’, it might turn out that majority of AAD cases were mild, transient and without a real impact on patient’s comfort. This needs to be emphasized more. 28. The authors focused on certain species effectiveness, however it is known that probiotic effects are not only species-, but also strain-dependent. Please comment on it in the discussion.

REVIEWER	Vargas, Ashley National Institute of Child Health and Human Development
REVIEW RETURNED	17-Nov-2020

GENERAL COMMENTS	This is a very clear, easy to follow systematic review and meta-analysis of probiotic use to prevent AAD. I have three concerns/requests: 1) I do not see a strength of evidence analysis. This is a must to understand how confidence the scientific community is in the results a systematic review, please complete this. 2) Please explain the scientific rationale for allowing “no treatment” as a reasonable control group. Consider excluding these studies on the basis of an inappropriate comparison group. 3) Did response vary by antibiotic name, type, route of administration, length or dose? Since probiotics would be preventing a side-effect of an antibiotic and since diarrheal response varies by antibiotic – I would highly encourage the authors complete some subgroup analysis with respect to the antibiotic used as they are able given the data available from each study. Minor comments: In the abstract please state the strength of evidence found (moderate?) Pg 4 line 52 – reword for clarity and consider – only certain species… were found to be effective Pg 12 line 33 – please state the strength of the evidence and update your methods regarding how you determined the strength of the evidence Pg 14 line 14 – did all studies attempt to capture and report on adverse events? If so, please state this, if not clarify how many studies did not attempt to capture or report on adverse events. Please add to the discussion the relevance of these findings to major medical association positions on use of probiotics including AGA (https://www.gastrojournal.org/article/S0016-5085(20)34729-6/fulltext)

REVIEWER	Goldenberg, Joshua National University of Natural Medicine, Helfgott Research Institute
REVIEW RETURNED	20-Nov-2020

GENERAL COMMENTS	This is a well written paper overall and the authors are to be commended. There are a few issues outstanding that should be addressed.

	Major issues: In applying a published quality 'checklist' to this review (AMSTAR2) this review would be rated at "Critically Low Quality". This is primarily due to issues in four critical domains: 1. Search: a grey literature was not searched, studies were limited to English and a justification for this is not given (AMSTAR2 item 4). 2. I did not see a list of excluded studies nor a justification for exclusions (AMSTAR2 item 7). 3. While the authors report on risk of bias using the Cochrane Risk of Bias tool they do not explore how this might impact the overall results of the study (AMSTAR2 item 13). In other words there is no sensitivity analysis using low risk fo bias studies only for example. 4. There does not appear to be an a priori protocol (AMSTAR2 item 2). Non critical domains: 1. I don't see where the authors reported on the sources of funding for the included studies (MSTAR2 item 10). Other issues: Publication bias: while the authors conducted a funnel plot visual assessment they did not conduct any formal statistical tests such as Eggers regression. This is not essential but could be added if the authors wish. The search is 'stale' it was conducted >1 year ago and should be updated and the overall results updated before publication. Heterogeneity is discussed but not explored. Subgroups have no statement or evaluation for credibility (e.g. ICEMAN) and should not be presented in the abstract unless very credible. There seems to be some confusion with Risk of Bias. The domains of "Selective outcome reporting" and "Missing outcome data" seem to be conflated.

REVIEWER	Abner, Sophia University of Leicester, College of Health Sciences
REVIEW RETURNED	04-Feb-2021

GENERAL COMMENTS	The aims of the systematic review and meta-analysis are clear and the statistical methods used for the meta-analysis are in line with the study objectives. Specifically, the method to use random effects where there is high heterogeneity is correct. The authors state they will use fixed effects where heterogeneity is low but the use of this has not been indicated. The authors have also considered the limitations of their study in the review in regards to the different strains of probiotic species, varying dosage amounts, and difference in comparison groups by performing subgroup analyses. I2 statistics have been provided along with odds ratios and their corresponding 95% confidence intervals and p-values, and a funnel plot. A sensitivity analysis was also performed. There are some minor comments the authors may wish to address. In two of the sub group analyses the I2 statistic has not been reported and the type of model used is not stated (probiotic dose and probiotic strain). Additionally it may be beneficial to include the results of the sensitivity analysis in the supplementary material and report the p-value of the funnel plot. Overall, the statistical methodology in this systematic review and meta-analysis has been well thought out and the limitations of their review have been addressed through using sub group analyses.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Prof. Hania SZAJEWSKA, The Medical University of Warsaw
Comments to the Author:
The manuscript 'Probiotics for the prevention of antibiotic-associated diarrhoea: A systematic review and meta-analysis' aims to update current evidence on this topic. The research question is clear and clinically important. The authors analysed a substantial amount of evidence and provided clear conclusions. There are, however, numerous issues which negatively impact the manuscript's credibility and consistency.

Reply: We would like to thank the Reviewer for their time and effort to review our work and provide feedback that increased the quality of our research output.

General comments
1. The authors decided to choose statistical model (i.e. fixed vs. random effects) depending on heterogeneity of the results. However, Cochrane Handbook clearly states that 'The choice between a fixed-effect and a random-effects meta-analysis should never be made on the basis of a statistical test for heterogeneity' (chapter 10.10.4).

Reply: We would like to thank the reviewer for this suggestion. We have updated our statistical analysis to use a random effects model due to significant heterogeneity between study participant age groups, probiotic species and doses and protocols implemented and difference in outcome measurement. This decision was purely based on the studies' methodological differences and not on statistical tests.

2. The authors state, that their main outcome was "incidence of AAD, classified as three or more episodes of loose stools (5-7 Bristol Stool Scale), in any 24-hour period during the antibiotic treatment or follow-up phases." Please clarify whether all included trials reported the outcome as stated above, including the use of Bristol Stool Scale. At least one study documented that there is a proven, large outcome heterogeneity in studies on AAD [doi: 10.1371/journal.pone.0228824].

Reply: We have added a sentence describing the heterogeneity of the definition of AAD used. Please see page 7 under "Subgroup and heterogeneity analysis".

3. The authors have chosen odds ratios as a method of presenting intervention effects. The Cochrane Handbook states that odds and odds ratios are more difficult to interpret for a reader. Please consider presenting the results as risk ratios.

Reply: We have updated our analysis to use risk ratios as all studies were RCTs and therefore prospective

4. Throughout the manuscript, the terms 'diarrhea' and 'AAD' are used interchangeably. Please standardize the terminology in this respect.

Reply: The terminology has now been standardised to the use of term AAD throughout the manuscript.

5. The term 'flora' used throughout the manuscript is historical, and should be replaced by 'microbiota'. Also, C. difficile no longer belongs to Clostridium genus, but to Clostridioides, and should be renamed in the manuscript accordingly.

Reply: We have performed these suggested amendments in the revised manuscript.

6. Throughout the manuscript, the authors refer to Joshua Goldenberg's 2013 systematic review as the 'most recent' Cochrane review. However, there was an update of this work by the same author in 2017. Please revise.

Reply: We would like to thank the reviewers for highlighting this issue. We have updated our reference to this Cochrane review throughout our review.

7. The database search was performed over a year ago. A search update is highly recommended.

Reply: We have updated the search and added seven studies to the review as requested. Changes are made throughout the updated manuscript (methods, results, graphs and tables).

Section-specific comments

Abstract:

8. "…co-administration of probiotics reduces the risk of antibiotic-associated diarrhoea in adults by 46% (Odds ratio (OR)=0.54 [95% CI: 0.44, 0.67], p<0.00001)" – risk and odds are two entirely different things. This sentence needs rephrasing.

Reply: The manuscript and the abstract have been revised to only use Risk Ratios. Please see updated abstract.

9. "Probiotics are non-pathogenic live micro-organisms, thought to improve the microbial balance of the host and reduce colonisation by pathogenic bacteria." – the definition presented by the authors is different from the widely adopted ISAPP definition (https://isappscience.org/for-scientists/resources/probiotics/). Please revise in the Abstract and Introduction.

Reply: We would like to thank the reviewer for the suggestion. The Abstract and Introduction have been revised as per Reviewer's suggestion. Please see pages 2 and 3.

10. ‘Design: Systematic review and meta-analysis.’ Please revise to precisely state ‘systematic review and meta-analysis of randomized-controlled trials’.
Introduction:

Reply: We have corrected this sentence.

11. “It is estimated that the increase in hospital admission due to AAD is 3.6 to 21.3 days” – hospital admissions cannot be measured in days, in contrast to hospital stay. Please revise.

Reply: We have corrected this sentence. Please see page 3, paragraph 4.

12. “They can be delivered orally, adhere to target gastrointestinal epithelium and are stable in acid and bile (Dunne et al. 2001).” – this statement seems to be a slight oversimplification of our current understanding of probiotics’ mechanism of action. In fact, in many cases this mechanism is unknown. According to ISAPP, ‘In some cases, probiotic mechanisms of action are known; in other cases, they are not even though a health benefit has been demonstrated. A common misconception is that probiotics need to alter the gut microbiota in order to be effective. In fact, probiotics are not known to take up permanent residence in the gut despite documented health benefits.’

Reply: We have revised this section. Please see page 3, paragraph 2.

13. ‘One of the most expensive probiotics in Australia costs less than four dollars daily (VSL#3 AU$3.96 per day, available from NourishMeOrganics.com.au, April 2020).’ – usually, use of commercial product names is discouraged in articles. Please refer to BMJ Open author guidelines on this subject.

Reply: We have updated the cost per day and omitted the commercial product information. Please see page 3, paragraph 4.

Methods:
14. The authors state, that “a sensitivity analysis was performed”. However, it is not described what kind of a sensitivity analysis it precisely was. Please clarify.

Reply: We have now discussed the sensitivity analysis further within the manuscript. Please see page 5, line 2; page 7, line 8; and page 9, paragraphs 2 and 3.

15. “Randomised controlled trials (RCTs) comparing probiotic use (any strain, dose or formulation) to placebo, alternative dose, alternative probiotic strain, or no treatment, for the prevention of diarrhoea in adults receiving antibiotic therapy were eligible for inclusion in this review.” There is a

discrepancy between this part and the abstract, which states that only 'placebo or no treatment control' will be taken into an account. Please clarify.

Reply: We have clarified this in the manuscript. Please see abstract and page 4, line 2.

Results:

16. "A total of 346 articles were identified in the initial search, of which 53 were eligible for full text review, and 37 of those articles were eligible for inclusion in the study." – if so, this search strategy has impressive specificity, considering that in recent Cochrane review mentioned previously, as much as 1351 studies were identified from the primary electronic databases. This surprisingly low number should be at least commented on in the discussion

Reply: One explanation for this lower number of articles identified in the initial search may be that the search was conducted as a title search, as opposed to keyword search. This was indicated in our a priori protocol. Please see supplemental file containing our protocol.

17. There are some missing data in Table 1, e.g. in column 2 of references 31-33, in column 5 of reference 32.

Reply: We have filled in missing data for reference 33 column 6, but were unable to find any other areas of missing data. This may have been a problem with a version of the manuscript you were viewing. If this is still the case please let us know. Please see revised table 1.

18. "All studies reported the outcome of interest; incidence of AAD as measured by three or more occurrences of diarrhoea in a 24-hour period". Firstly, I think that this sentence should read "occurences of loose stool" rather than "occurences of diarrhoea". Secondly, in reference to my previous comment – did really all those authors use Bristol Stool Scale? Haven't you identified at least one trial which used different definition of diarrhoea, or didn't use any definition at all? Or perhaps, did you exclude trials which didn't provide a precise definition of diarrhoea?

Reply: We would like to thank the Reviewer for raising this important point. We have added a column to revised Table 1 indicating how each study measured diarrhoea.

19. "Overall 51% of studies were rated as having a low risk of bias, while 49% were at unclear risk of bias." What were the criteria for a study to be ranked as "low" or "high" risk of bias?

Reply: We have used the Cochrane risk of bias tool. We have presented the complete risk of bias summary for readers to see the review authors' judgement about each risk of bias item for each included study. Please see figure 2.

20. "…demonstrated a statistically significant reduction in the incidence of AAD by 46% (Odds ratio

(OR)=0.54 [95% CI: 0.44, 0.67], p<0.00001).” – similar comment as before – incidence and odds are two different things. Please revise throughout the manuscript for consistency.

Reply: We have only used Risk Ratios in the revised manuscript.

21. “This suggests that the number needed to treat (NNT), to prevent one case of AAD would be 19 patients.” – how was NNT calculated? On the basis of OR? If so, then according to Cochrane Handbook, the authors should provide an information and rationale on the ‘assumed comparator risk’ (chapter 15.4.4.).

Reply: We have revised our calculation of NNT to be based on raw data using the online calculator found at https://clincalc.com/Stats/NNT.aspx.

22. “On inspection, the funnel plot of the complete case analysis (Figure 3) shows little evidence of publication bias. This suggests that our results are robust to theoretical large future trials with results in the opposite direction (i.e. that probiotics increase the risk of AAD rather than decreasing it).” – I don’t understand, how low risk of publication bias translates to robustness of results to future studies with different results. Also, visually, the funnel plot is not necessarily symmetric – there are six small studies (SE>1) with results more favourable towards probiotics compared to only 2 small studies on the right side of the plot. Did you consider performing tests for asymmetry?

Reply: Since updating the search we have added additional studies and updated the funnel plot. On consultation with our statistician the symmetry of the funnel plot was found to be acceptable.

23. “No serious adverse events were reported in any of the included studies” – it would be also useful to include information, how many studies reported adverse events at all.

Reply: We have included this information in the results section. Please see page 9, paragraph 4.

Discussion

24. The results should be briefly recapped rather than rewritten completely with p-values, confidence intervals etc. Please revise.

Reply: We have updated the discussion section to reflect the above advice. Please see page 9-10.

25. Some information regarding the results is presented for the first time, eg. “we found significant heterogeneity between study designs and population, severity of illness of the study group, as well as probiotic species used, dose, timing and duration of probiotic administration.”, or “In the five RCTs investigating adjunct probiotic use with antibiotics for H.pylori treatment we found a reduction in diarrhoeal side-effects (OR=0.43, [95% CI: 0.27, 0.69], p<0.001).”. Data like this should be introduced in the “results” section, and only recapped or commented on in the discussion.

Reply: We have now introduced both of the above points in the results section (page 7, paragraph 6 and page 9, paragraph 1) and only commented on them in the discussion.

26. "One study [62] compared two strains of L. casei, Actimel® (L. casei DN-114001) and Yakult® (L. casei Shirota), for the prevention of AAD. In this study L. casei DN-114001 was significantly more effective than L. casei Shirota , however, we were unable to include this data in our review as it was not randomised." – I don't understand, why this specific study is mentioned? I don't think that it contributes much to the discussion. I guess there were more studies excluded for similar reasons, so why mention this one?

Reply: We mentioned this study because it was the only one we found that made a direct comparison of strains. We have revised our discussion of this point. Please see page 10, paragraph 5.

27. "A limitation of this study is that the only outcome measured was incidence of diarrhoea. This review's utility in clinical practice may have been improved by also analysing duration of AAD." – this is a very important point, however I would also add for consideration: diarrhea severity, need for hospitalization due to diarrhoea, quality of life and other outcomes of real-life importance. Considering the fact, that the definition of diarrhoea in the included trials was '3 or more loose stools in 24 h', it might turn out that majority of AAD cases were mild, transient and without a real impact on patient's comfort. This needs to be emphasized more.

Reply: We agree with the Reviewer and we have updated our limitations paragraph to discuss this point. Please see page 10, paragraph 7.

28. The authors focused on certain species effectiveness, however it is known that probiotic effects are not only species-, but also strain-dependent. Please comment on it in the discussion.

Reply: We would like to thank the Reviewer for highlighting this. We have updated the discussion accordingly. Please see page 10, line 1.

Reviewer: 2
Dr. Ashley Vargas, National Institute of Child Health and Human Development
Comments to the Author:
This is a very clear, easy to follow systematic review and meta-analysis of probiotic use to prevent AAD. I have three concerns/requests:

1) I do not see a strength of evidence analysis. This is a must to understand how confidence the scientific community is in the results a systematic review, please complete this.

Reply: Thank you for your comment. We have now performed a strength of evidence analysis using the GRADE criteria. Please see page 4, paragraph 5 and page 9, paragraph 6.

2) Please explain the scientific rationale for allowing “no treatment” as a reasonable control group. Consider excluding these studies on the basis of an inappropriate comparison group.

Reply: The authors have discussed this point and feel that no treatment is an appropriate control as this would be considered “usual care” in clinical practice.

3) Did response vary by antibiotic name, type, route of administration, length or dose? Since probiotics would be preventing a side-effect of an antibiotic and since diarrheal response varies by antibiotic – I would highly encourage the authors complete some subgroup analysis with respect to the antibiotic used as they are able given the data available from each study.

Reply: More than half of the included studies did not list which antibiotics were administered, so we were unable to delineate this. Future studies could focus on treatments of specific antibiotics for this purpose. Please see page 11, paragraph 1.

Minor comments:
In the abstract please state the strength of evidence found (moderate?)

Reply: We added this in the abstract. Please see revised abstract.

Pg 4 line 52 – reword for clarity and consider – only certain species… were found to be effective

Reply: The sentence was reworded.

Pg 12 line 33 – please state the strength of the evidence and update your methods regarding how you determined the strength of the evidence

Reply: We have now performed a strength of evidence analysis using the GRADE criteria. Please see page 4, paragraph 5 and page 9, paragraph 6.

Pg 14 line 14 – did all studies attempt to capture and report on adverse events? If so, please state this, if not clarify how many studies did not attempt to capture or report on adverse events.

Reply: We have included this information in the results section. Please see page 9, paragraph 4.

Please add to the discussion the relevance of these findings to major medical association positions on use of probiotics including AGA (https://www.gastrojournal.org/article/S0016-5085(20)34729-6/fulltext)

Reply: We added this information in the revised discussion. Please see page 11, paragraph 5.

Reviewer: 3
Dr. Joshua Goldenberg, National University of Natural Medicine
Comments to the Author:
This is a well written paper overall and the authors are to be commended. There are a few issues outstanding that should be addressed.

Major issues: In applying a published quality 'checklist' to this review (AMSTAR2) this review would be rated at "Critically Low Quality". This is primarily due to issues in four critical domains:

1. Search: a grey literature was not searched, studies were limited to English and a justification for this is not given (AMSTAR2 item 4).

Reply: We have now performed a grey literature search. Studies included were limited to English because our group only speaks English. Justification was provided in the revised manuscript. Please see page 4, line 3.

2. I did not see a list of excluded studies nor a justification for exclusions (AMSTAR2 item 7).

Reply: As this was not a Cochrane review, we were not required to provide a list of excluded studies, with justifications for their exclusions, within the submission to BMJ Open. However, this information is held within the Covidence online software and summarised in the PRISMA flow chart. Please see Figure 1.

3. While the authors report on risk of bias using the Cochrane Risk of Bias tool, they do not explore how this might impact the overall results of the study (AMSTAR2 item 13). In other words there is no sensitivity analysis using low risk fo bias studies only for example.

Reply: We would like to thank the Reviewer for highlighting this point. We have now included a sensitivity analysis. Please see page 7, paragraph 2 and page 9, paragraphs 2 and 3.

4. There does not appear to be an a priori protocol (AMSTAR2 item 2).

Reply: There was an a priori protocol which was submitted and reviewed within the medical school. We will include it as a supplementary file.

Non critical domains: 1. I don't see where the authors reported on the sources of funding for the included studies (MSTAR2 item 10).

Reply: This was discussed by the authors while making their judgement of “other sources of bias” within the risk of bias assessment.

Other issues:
Publication bias: while the authors conducted a funnel plot visual assessment they did not conduct any formal statistical tests such as Eggers regression. This is not essential but could be added if the authors wish.

Reply: Since updating the search we have added additional studies and updated the funnel plot. On consultation with our statistician the symmetry of the funnel plot was found to be acceptable and it was not felt that any formal statistical tests would provide any further significant information.

The search is 'stale' it was conducted >1 year ago and should be updated and the overall results updated before publication.

Reply: We have updated the search and added seven studies to the review as requested. Changes are made throughout the updated manuscript (methods, results, graphs and tables).

Heterogeneity is discussed but not explored.

Reply: We have explored heterogeneity, both statistical and methodological, and discussed their effects. On this basis we have now chosen to use a random effects model for all analyses.

Subgroups have no statement or evaluation for credibility (e.g. ICEMAN) and should not be presented in the abstract unless very credible.

Reply: As per another reviewer’s comment we have now performed a strength of evidence analysis using the GRADE criteria. Please see page 4, paragraph 5 and page 9, paragraph 6.

There seems to be some confusion with Risk of Bias. The domains of "Selective outcome reporting" and "Missing outcome data" seem to be conflated.

Reply: Thank you for highlighting this issue. We have reviewed this section and made appropriate changes in the manuscript.

Reviewer: 4
Dr. Sophia Abner, University of Leicester
Comments to the Author:
The aims of the systematic review and meta-analysis are clear and the statistical methods used for the meta-analysis are in line with the study objectives.

Specifically, the method to use random effects where there is high heterogeneity is correct. The authors state they will use fixed effects where heterogeneity is low but the use of this has not been indicated.

Reply: Following Reviewers' advice, we have decided to use a random effects model for all analyses as there is significant heterogeneity between study methodologies. Please see changes throughout the manuscript.

The authors have also considered the limitations of their study in the review in regards to the different strains of probiotic species, varying dosage amounts, and difference in comparison groups by performing subgroup analyses. I2 statistics have been provided along with odds ratios and their corresponding 95% confidence intervals and p-values, and a funnel plot. A sensitivity analysis was also performed.
There are some minor comments the authors may wish to address. In two of the sub group analyses the I2 statistic has not been reported and the type of model used is not stated (probiotic dose and probiotic strain).

Reply: We would like to thank the reviewer for the time they spent to provide productive feedback. We have revised our discussion of the I2 statistic in the manuscript and a random effects model was used for all analyses due to the heterogeneity of study methodology. Please see page 7, paragraphs 4 and 6.

Additionally it may be beneficial to include the results of the sensitivity analysis in the supplementary material and report the p-value of the funnel plot.

Reply: We have included these results in the revised manuscript's supplementary material as per the Reviewer's suggestion.

Overall, the statistical methodology in this systematic review and meta-analysis has been well thought out and the limitations of their review have been addressed through using sub group analyses.

Reply: We would like to thank the Reviewer for their kind words.

VERSION 2 – REVIEW

REVIEWER	Vargas, Ashley National Institute of Child Health and Human Development
REVIEW RETURNED	16-Jul-2021

GENERAL COMMENTS	Well done, this was a major overhaul and a lot of work. This paper is excellent, has addressed all my concerns at this point and provides a valuable addition to the literature.